ADVENTIST UNIVERSITY OF CENTRAL AFRICA (AUCA)

# Distributed Multi-Model Analytics for E-commerce Data

Individual Big Data Analytics Final Project

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Technical report | MongoDB, Apache HBase, Apache Spark, integrated analytics and visualization

Evidence basis: local execution outputs and screenshots supplied during project development. Figures reproduced from reported numeric results where the original image files were not available to this report generator.

# 1. Executive summary and objectives

This project demonstrates a distributed, multi-model analytics workflow for synthetic e-commerce data. Document-oriented records are analysed with MongoDB, user activity and daily product metrics are stored in Apache HBase, and Apache Spark is used for batch processing, SQL joins and an acquisition-to-purchase analysis. The goal is to combine storage models suited to different access patterns rather than force all data into one database.

The demonstrated sample includes 10,000 sessions. Spark identified 1,001 sessions linked to purchases, giving an actual purchase conversion rate of 10.01%. MongoDB category analysis ranked cat_019 (Powell PLC) first with $126,677.59 in line-item revenue; customer analysis ranked user_000104 first with $12,034.07 in total spending. HBase successfully stored and retrieved event records and daily metrics.

## Project objectives

1. Generate structured but varied synthetic users, products, categories, transactions and web sessions.
2. Implement and query document and wide-column data models.
3. Execute Spark batch and SQL analytics across related datasets.
4. Analyse conversion and revenue, and communicate findings visually.

## 2. System architecture

| Layer | Role and evidence |
|---|---|
| Synthetic JSON generator | Creates users, products, categories, transactions and sessions as JSON files. |
| MongoDB | Stores analytical collections and supports category revenue and customer spending aggregations. |
| Apache HBase 2.1.3 in Docker | Stores user_events and product_daily_metrics for keyed retrieval. |
| Apache Spark / PySpark | Reads JSON-derived DataFrames, performs joins and aggregations, and computes the purchase funnel. |
| Python / Matplotlib | Produces category revenue, session status and referral conversion charts. |

The local deployment is a proof of concept: Spark runs in local mode and HBase runs in a single Docker container. These choices support functional validation but do not demonstrate multi-node throughput or fault tolerance.

# 3. Dataset and data modelling

The generated dataset includes five core JSON entities. Users contain customer attributes; products link to categories; categories provide category names; transactions contain user/session identifiers, timestamps and arrays of purchased items; and sessions contain page views, viewed products, cart contents, referral channels and conversion status. The generator was executed locally and produced files named users.json, products.json, categories.json, transactions.json and sessions_0.json.

## MongoDB modelling

MongoDB is suited to transactions with nested line items and sessions with variable-length behavioural arrays. The project contains an import script (mongodb/import_data.py) and two analytical scripts (mongodb/category_revenue.py and mongodb/customer_analysis.py). Both analyses were run successfully against the local database. The category query groups completed transaction items by product category; the customer query ranks spending and order counts by user.

## HBase row-key design

| Table | Column families | Observed access pattern |
|---|---|---|
| user_events | event, session | User/event-oriented row keys; point lookup of an event record. |
| product_daily_metrics | metrics | Product/date-oriented row keys; scan or retrieve daily product view metrics. |

The HBase shell confirmed that both tables were enabled. A sample event row used key user_test001#20261110#001 with event:type=product_view, event:product_id=prod_001 and session:session_id=session_test001. A get operation returned all three values. Subsequent counts showed 365 rows in user_events and 360 rows in product_daily_metrics. A sample product metrics scan returned metrics:views=1 for several product/date keys.

## Data-quality considerations

The generated data is synthetic, so category names and country codes should not be interpreted as a real merchant portfolio or real customer geography. A session labelled converted does not necessarily have a linked transaction; the funnel analysis therefore distinguishes status labels from verified transaction linkage.

# 4. MongoDB analytics: verified results

## 4.1 Top product categories by revenue

| Category | Revenue (USD) | Units | Transactions |
|---|---|---|---|
| cat_019 / Powell PLC | 126,677.59 | 559 | 262 |
| cat_013 / Smith Ltd | 112,609.70 | 424 | 196 |
| cat_007 / Moore, Becker and Carlson | 103,383.65 | 330 | 155 |
| cat_024 / Patterson, Smith and Jones | 100,004.50 | 320 | 158 |
| cat_018 / Phillips Inc | 96,699.11 | 324 | 160 |

The leading category generated approximately $14,068 more revenue than the second-ranked category. Revenue is calculated from completed transaction line items using quantity multiplied by unit price. These are synthetic monetary values in the dataset, not audited business sales.

## 4.2 Highest-spending customers

| Customer | Country code | Total spend (USD) | Orders |
|---|---|---|---|
| user_000104 | VE | 12,034.07 | 8 |
| user_000367 | MD | 11,850.46 | 7 |
| user_000496 | RO | 11,841.00 | 7 |
| user_000210 | WS | 11,788.22 | 11 |
| user_000127 | KG | 10,767.88 | 7 |

Customer user_000104 led total spending, with eight orders and an average reported order value of $1,504.26. Customer rankings provide a basis for segmentation, although no causal claims about loyalty or marketing effectiveness can be made from synthetic records.

## Query validation

The two scripts completed without error in the supplied Command Prompt output and returned ranked results. The final repository should retain the scripts and document the database connection configuration and loading procedure for reproducibility.

# 5. Spark processing and integrated analytics

## 5.1 Spark SQL category analysis

Spark SQL joined transactions and nested items to products and categories to rank categories by gross line-item revenue, units sold and transaction counts. The top result was cat_019 (Powell PLC) with $126,677.59, 559 units and 262 transactions. The agreement with the MongoDB ranking provides a useful cross-check of the two analytical implementations.

## 5.2 Session-to-purchase funnel

| Metric | Verified value |
|---|---|
| Sessions analysed | 10,000 |
| Sessions linked to a purchase | 1,001 |
| Actual purchase conversion rate | 10.01% |
| Linked purchase revenue (USD) | 791,391.30 |
| Sessions labelled converted | 2,009 |
| Sessions labelled browsed | 3,429 |
| Sessions labelled abandoned | 4,562 |

The integrated analysis links sessions with transactions to count observed purchasing sessions. Only 1,001 of the 2,009 sessions carrying a converted status were linked to an actual purchase in the reported analysis. Consequently, status-based conversion (20.09%) and transaction-linked purchase conversion (10.01%) must not be presented as interchangeable metrics.

## 5.3 Referral channel comparison

| Channel | Sessions | Purchases | Purchase rate |
|---|---|---|---|
| Email | 2,036 | 219 | 10.76% |
| Affiliate | 2,013 | 208 | 10.33% |
| Social | 1,952 | 195 | 9.99% |
| Direct | 2,009 | 197 | 9.81% |
| Search engine | 1,990 | 182 | 9.15% |

Email had the highest observed purchase conversion rate, while search engine had the lowest. These descriptive differences do not establish statistically significant channel effects, and the synthetic data should not be used to recommend real advertising expenditure.

# 6. Visualization and interpretation

Three PNG charts were generated successfully on the student computer by visualizations/create_charts.py: category_revenue.png, session_status.png and referral_conversion.png. The following tables summarize their underlying interpretation; the original PNG files should be included in the GitHub repository and can be inserted into a final report revision.

### Figure 1. Category revenue ranking




Reconstructed from the five leading MongoDB/Spark category values shown in execution output.

### Figure 2. Session status distribution




Status labels are generator fields; these are not the verified purchase counts.

### Figure 3. Purchase conversion by referral channel




This figure uses verified transaction-linked purchase rates from Spark, rather than the separate status-based referral chart.

# 7. Scalability, limitations and conclusion

## Scalability considerations

MongoDB can support indexing and aggregation over nested commerce documents; HBase supports sparse, wide-column records with row-key-oriented retrieval; and Spark distributes batch transformations across partitions when deployed on a cluster. A production implementation would require measured partitioning strategy, suitable HBase row-key distribution, database indexes, resource monitoring and a repeatable ingestion pipeline. The present tests validate functionality locally, not performance at production scale.

## Limitations

First, the dataset is synthetic and its category names, referral behaviour and customer spending are generated rather than observed. Second, the Spark environment was tested in local mode and HBase was deployed in one container, so no cluster scalability benchmark was performed. Third, status-based session conversion and purchase-linked conversion diverge, illustrating the need for a precise business definition. Fourth, the report is based on console evidence provided during development; full reproducibility requires the final repository, environment details and original output files.

## Conclusion

The project demonstrates a functioning multi-model analytics pipeline: MongoDB analyses successfully ranked category revenue and customer spending; HBase stored and returned user events and product/day metrics; Spark SQL computed category performance; and integrated Spark analysis measured 1,001 purchasing sessions from 10,000 sessions (10.01%). The three generated charts provide a visual summary of category revenue, session status and referral behaviour. The most important analytical lesson is to verify purchase events rather than assume a converted status implies a completed transaction.

## Reproducibility and submission checklist

Before submission, verify the GitHub repository includes the JSON generator, MongoDB import and aggregation scripts, HBase schema/loading scripts, Spark processing/SQL/funnel scripts, visualization script, generated charts and README. The README should specify installation prerequisites, run order, database configuration and expected outputs. Capture evidence of HBase counts and retrieval, MongoDB aggregations, and Spark execution. Confirm the repository link is accessible to the examiner.

## Implementation references

Project brief: AUCA Final Project Oct 2026.pdf (provided course assignment). Local source paths observed: data/dataset_generator_ready.py; mongodb/import_data.py, category_revenue.py, customer_analysis.py; visualizations/create_charts.py; Spark analysis scripts and HBase loading scripts within the corresponding project folders. Exact dependency versions and the final GitHub URL should be confirmed from the submitted repository.